Adán Daniel Márquez Hernández

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EXPERIENCE

Model Based Controls Engineer at Daikin (Monterrey, N.L.) July 2025 - Present

- Implemented an Azure DevOps pipeline to deliver MATLAB/Simulink code generation code to Software teams.
- Develop a Simulink Test Framework to validate control systems and algorithms based on requirements.
- Tested the control architecture in Model and Software in the Loop simulations.

Software Intern at VisionariesHub. (Remote, Monterrey, N.L.) March 2025 - July 2025

- Incorporate custom payment methods in production into a Wix web page with JavaScript.

Researcher at Tec. of Mty. (Monterrey, N.L.) August 2023 - June 2025

- Developed a novel quaternion extended state observer for attitude tracking and state estimation in Simulink.
- Implemented state estimation algorithms in Simulink/Simscape with model-based design for a multirotor.
- Executed multiple nonlinear observer-based controllers in Simulink/Simscape with Model and Software in the Loop.
- Established a deployment framework from Simulink to flight computers with embedded coder in MATLAB/Simulink.

Research Intern at Tec. of Mty. (Monterrey, N.L.) February-June 2023

- Implemented an adaptive nonlinear control scheme for a hexacopter under external bounded perturbations.
- Developed a controller with model-based design and Model in the Loop simulations in Simulink and Simscape.

Research Intern at Tec. of Mty. (Monterrey, N.L.) August - December 2022

- Designed and built an aerial gripper manipulator with SolidWorks, capable of lifting rectangular boxes of 200 g.
- Formulated and programmed a robust non-linear PD controller for a hexacopter in Simulink and ROS/Gazebo.

Robotic Cell Design - *Client: ABB* February - March 2022

- Developed a hydraulic clamp system for securing parts in a robotic welding cell using SolidWorks.
- Proposed and validated an autonomous robot trajectory for a welding process using ABB RobotStudio.

Linear Actuator Design and Construction - *Client: John Deere* October - December 2021

- Devised and built a low-cost linear actuator mechanism in Solidworks.
- Incorporated a closed-loop PID and on/off position controllers in Arduino and STM32 microcontrollers.

LEADERSHIP EXPERIENCE AND ACTIVITIES

ULEAD Bootcamp – *Client: Arroyo Vivo* January - February 2023

- Addressed and evaluated the pollution problem in the community “Campana Altamira”.
- Established a sustainable solution for the trash situation with a multidisciplinary team in a short period.

Programming Lead - *MRL: Monterrey Rocket Lab* August 2022 - April 2024

- Proposed an extended Kalman filter to fuse multiple sensors in a model rocket for apogee detection in MATLAB.

SKILLS & INTERESTS

Technical: Python, MATLAB, Simulink, C++, Linux, Git, Docker, Azure DevOps, Simulink Test.
Languages: English, French, German.
Interests: Origami, Book Binding, Running, and Muay Thai.

EDUCATION

Tecnológico de Monterrey, Campus Monterrey — B.S. in Mechatronics Engineering, GPA: 96/100

Tecnológico de Monterrey, Campus Monterrey — M.Sc. in Engineering., GPA: 96/100